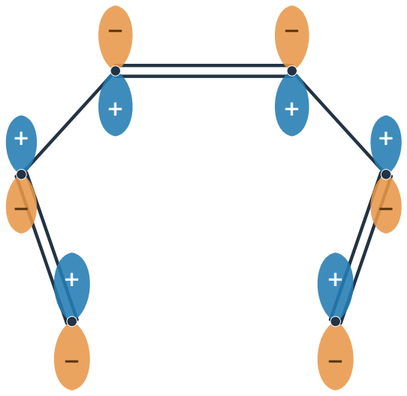
6π electrocyclic closure | orbital phases along the reaction

THERMALLY ALLOWED

disrotatory

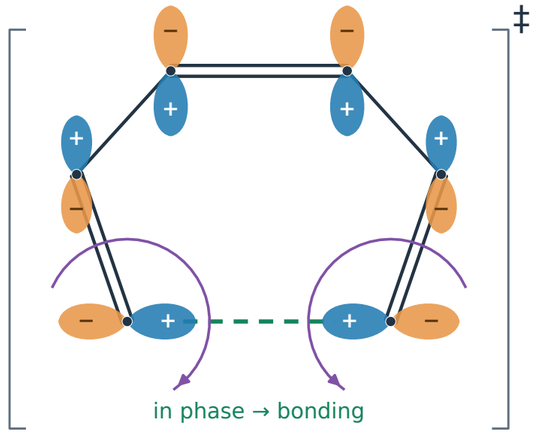
REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Polyene HOMO, ψ₃ (occupied)	Hückel circuit · 6 electrons	Product connectivity



Same terminal phases

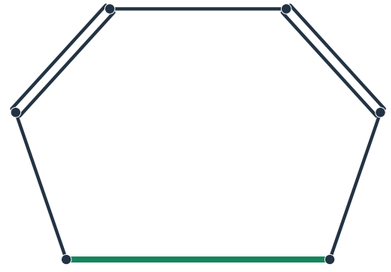
Follow the terminal lobes during rotation.

Δ



opposite rotation senses

Δ



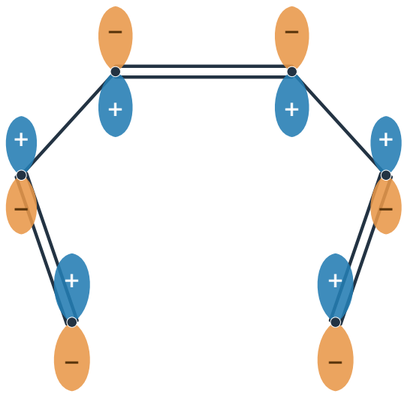
Cyclohexa-1,3-diene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

conrotatory

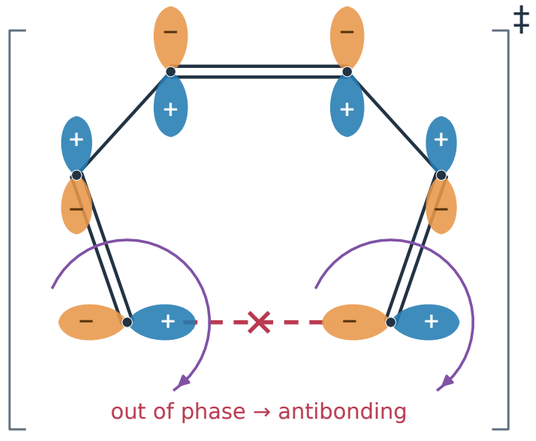
Polyene HOMO, ψ₃ (occupied)	Möbius circuit · 6 electrons	Proposed product connectivity



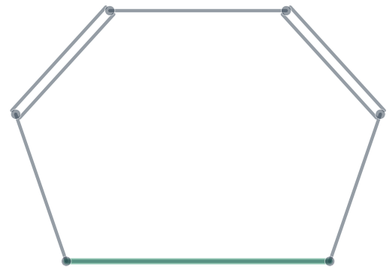
Same terminal phases

Follow the terminal lobes during rotation.

Δ



same rotation sense



Cyclohexa-1,3-diene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Rotation arrows follow the same viewing convention at both termini. Product connectivity is shown without substituent stereochemistry.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.